1. Weakest supply wins
A facility's level is the lowest of its responses across everything it needs, times its condition (1 if intact, 0 if damaged and not yet repaired). Full power does not make up for missing water.

Value
none

Example
B needs power and water. Power arrives in full. Water arrives at 50%, and B throttles on water. B runs at 0.5.

2. Throttle, trip, floor
When a supply falls short, the facility responds in one of three ways (shown on its card):
- ◐ throttle: runs at the share it receives;
- ⚡ trip: runs fully while it receives at least the trip point, otherwise shuts off;
- ▁ floor: a hospital whose power falls short runs at its essential level (0.6) on its generator, as long as the generator runs and has fuel. When the fuel runs out, it responds like a trip: it runs fully while it receives at least the trip point, otherwise it shuts

2. Throttle, trip, floor
(cont. 2/2)
off.

Value
trip point 0.9; floor 0.6

Example
Supply arrives at 70%. Throttle: 0.7. Trip: 0 (70% is below the trip point). Floor: 0.6.

3. How supply passes along an arrow
Each arrow on the board has a form:
- linear: it passes κ (the *share*) × the supplier's level;
- threshold: it passes κ in full while the supplier is at or above θ (the *cut-off*), otherwise nothing.

κ is the share of the user's need that this supplier covers. Telecom arrows work differently: see card 5.

Value
threshold θ 0.5

3. How supply passes along an arrow
(cont. 2/2)
Example
A is at 0.8 and supplies B along a linear arrow with κ 0.5. B receives 0.4.

4. More than one supply
Within its own network (a substation fed by two lines), a facility takes the largest supply arriving, up to full. Losing one line costs nothing if another still carries full supply.

From another network, supplies come in feed groups (on the card: single, two half, two full). Each group passes the smallest of its arrows (a group needs all its arrows); the groups add up, to at most full.
- two half: each feed covers half, so losing one leaves 0.5.

4. More than one supply
(cont. 2/2)
- two full: either feed covers it all, so losing one leaves 1.

Value
none

Example
B is fed two-half by suppliers S1 and S2. S1 goes out. B receives 0.5.

5. When telecom is lost
Facilities can work for a while without telecom, by phone, paper and hand. Each card shows ◷ T (hours of full function) and d_eff (the level the missing telecom counts as afterwards).
- Telecom arrows pass the supplier's level without κ. On a telecom arrow, κ is the share of the facility's telecom that runs over the public network; it is built into d_eff = 1 − κ(1 − d).
- When telecom falls short, the facility runs fully for T hours.
- After T hours, the telecom it receives counts as r + (1

5. When telecom is lost
(cont. 2/3)
− r) × d_eff, where r is the share still arriving. Its telecom response then acts on that: throttle runs at it; trip runs only if it is at least the trip point (0.9), and otherwise stops.
- The T-hour clock starts again from zero each time telecom is fully back.

Value
T, d and κ from each facility's card

Example
C loses all telecom at hour 0; T = 6 h, d_eff = 0.7. If C throttles on telecom, it runs

(see the other side)

2. Throttle, trip, floor
Drill
C trips on power. Power arrives at 92%. What level is C at? What if it arrives at 89%?

Answer
At 92%, 1 (at or above the trip point). At 89%, 0.

1. Weakest supply wins
Drill
B needs power (response: throttle) and water (response: trip). Power arrives at 100%. Water arrives at 85%. What level is B at?

Answer
0. Water is below the trip point (0.9), so the water response is 0, and the lowest response sets B's level.

4. More than one supply
Drill
Substation B is fed within its network by lines from A1 and A2. The line from A1 fails; A2 is intact. What does B receive? Then: a pump fed two-full by two substations loses one of them. What does it receive?

Answer
B receives 1 (the largest supply arriving). The pump receives 1 (the other group covers it all).

(see the other side)

3. How supply passes along an arrow
Drill
A is at 0.6 and supplies B along a threshold arrow with κ 1. What does B receive? What if A drops to 0.4?

Answer
At 0.6, 1 (A is at or above θ). At 0.4, 0.

(see the other side)

5. When telecom is lost
Drill
B throttles on telecom and loses it at hour 0. Its card shows T = 12 h and d_eff = 0.9. Telecom returns at hour 20. What is B's level at hours 6, 15 and 21?

Answer
1 at hour 6; 0.9 at hour 15; 1 at hour 21.

(see the other side)

5. When telecom is lost
(cont. 3/3)
at 1 until hour 6, then at 0.7. If C trips on telecom, it runs at 1 until hour 6, then stops (0.7 is below the trip point).

6. Stocks drain and refill
▮ A stock covers its own supply while it lasts: a battery or generator covers power, a water tank covers water. The card shows hours at full draw. A hospital on its generator draws only its essential load, so its fuel lasts longer than the card's figure.

A generator that fails to start never covers. One that fails to run stops at its time on the generator card. Battery and generator hours add.

Refill once the supply returns:

6. Stocks drain and refill
(cont. 2/3)
- battery: empty to full in 12 h;
- water storage: from the pumps' spare capacity;
- purified water, make-up water and hospital water: from the main, at 1 × the rate it drained;
- generator fuel: only by delivery (fuel card);
- national fuel stocks: never, inside the 72 hours.

Value
battery recharge 12 h; main refill 1 × drain rate

Example
Facility C, with a 5 h

6. Stocks drain and refill
(cont. 3/3)
battery, loses power at hour 0. It goes out at hour 5 unless power returns first.

7. Fuel deliveries
A generator's tank is refilled only by truck. The first delivery arrives 12 h after the facility loses power; then one every 24 h. Each delivery carries 24 h of fuel. Deliveries come only while the fuel depot is working and a road reaches the facility.

Value
lead time 12 h; interval 24 h

Example
Facility C, with 20 h of generator fuel, loses power at hour 0. Its first delivery arrives at hour 12, before the tank is empty.

8. A crew's job
For each job a crew:
1. travels: by road at 40 km/h, then off road at 2 km/h to a line;
2. prepares on site for 2 h (every job);
3. repairs once the parts are in. Parts lead time runs from hour 0, in parallel, with no crew.

A line is one job; its failure points are worked in turn, 6 h each, or 72 h at a collapsed tower. A repaired element is back in full.

Value
preparation 2 h; 6 h per

8. A crew's job
(cont. 2/2)
failure point; 72 h per tower

Example
A crew leaves at hour 0, travels 1.5 h, prepares 2 h, and repairs a line with one failure point in 6 h. The line is back at hour 9.5.

9. Which job a crew takes next
A free crew takes the waiting job that comes first in this order:
1. towered power lines and transmission substations;
2. distribution substations and power feeds to end uses, highest weight first (a substation counts the highest weight among what it supplies, its own included);
3. everything else.

Ties go to the shortest trip from where the crew stands, then the lowest id. Crews never go back to base between jobs.

9. Which job a crew takes next
(cont. 2/2)
Value
none

Example
A damaged pumping station close by and a transmission substation farther away are both waiting. The crew takes the transmission substation.

(see the other side)

6. Stocks drain and refill
Drill
B has a 4 h battery. Power is lost at hour 0 and returns at hour 3. Does B go out? When is its battery full again?

Answer
B never goes out. It used 3 of 4 hours, and refills at a full charge per 12 h, so it is full again at hour 12.

(see the other side)

8. A crew's job
Drill
A crew is assigned at hour 0 to a facility 1 h away. Its parts lead time is 5 h and the crew work is 8 h. When is the facility back?

Answer
Arrives at hour 1, prepared at hour 3, parts in at hour 5, so work runs 5–13. Back at hour 13.

7. Fuel deliveries
Drill
B's generator holds 10 h of fuel. B loses power at hour 0, and the power stays off. Is B ever out, and when?

Answer
Out from hour 10 to hour 12, when the first delivery arrives. From then on each delivery lands before the tank empties.

(see the other side)

9. Which job a crew takes next (cont. 2/2)
though it is closest.

9. Which job a crew takes next
Drill
Two crews are free at the base. Three jobs wait: a damaged pumping station 1 h away, a distribution substation feeding a hospital (weight 10) 3 h away, and a distribution substation feeding a clinic (weight 3) 2 h away. Which job does each crew take first?

Answer
Crew 1 takes the substation feeding the hospital (group 2, highest weight). Crew 2 takes the substation feeding the clinic (group 2). The pumping station waits (group 3),

(see the other side)

10. Travel between jobs
A crew that finishes a job goes straight to its next job from where it stands; it never returns to base. Its trip:

- from a facility to the nearest road, and from the road to a facility: driven;
- from a point on a line to the nearest road, and back on: walked at 2 km/h;
- along the roads in between: driven at 40 km/h;
- if no road joins the two places: straight across at road speed, plus 12 h.

Preparation (2 h) comes once per job, at its first site. Between failure points of the

10. Travel between jobs
(cont. 2/2)
same line, the crew walks along the line at 2 km/h, without preparing again.

Value
road 40 km/h; off road 2 km/h

Example
A crew finishes a job at a facility on the road at hour 5. Its next job is a line whose first failure point lies 4 km farther along the road and 1 km off it. The crew drives 4 km (0.1 h), walks 1 km (0.5 h), arrives at hour 5.6, prepares until hour 7.6, and starts the work.

11. Re-energizing by hand
Substations are switched remotely over telecom. If a repair should bring power back to a substation whose telecom is down, the power stays off until a crew travels there and switches it by hand: 1 h on site. The trip is a job like any other, at the same priority as the repair that triggered it. It is not needed when the repair is at the substation itself.

Value
1 h on site

Example
A line into substation S is

11. Re-energizing by hand
(cont. 2/2)
repaired at hour 10, but S's telecom is out. A crew 0.5 h away goes there; power returns at hour 11.5.

12. Time, steps and hours out
The model moves in steps of 15 minutes. Log every event at its exact time. A repair, a switching trip or a fuel delivery takes effect at the first step boundary at or after its exact time: one finished at hour 9.40 counts from 9.50; one finished at exactly 9.50 counts from 9.50. Crews are not rounded: a crew starts its next trip the moment it finishes. A facility's level holds for the whole step it is set in.

Hours out: a facility at level s for h hours counts $(1 - s) \times h$ hours out. Only end-use

12. Time, steps and hours out
(cont. 2/2)
facilities are scored; their weights are on their cards. The horizon is 72 h.

Value
step 15 min; horizon 72 h

Example
A repair finishes at hour 9.40. The model applies it at hour 9.50.

11. Re-energizing by hand
Drill
The only line into substation S is repaired at hour 20. S's telecom is working. When does S get power back? And if S's telecom were down, with a free crew 1 h away?

Answer
With telecom working: hour 20. With telecom down: hour 22 (20 + 1 h travel + 1 h on site).

(see the other side)

10. Travel between jobs
(cont. 2/2)
km along the line on foot, 1.0 h, so there at 12.8; no new preparation; done at 18.8.

12. Time, steps and hours out
Drill
An end-use facility is out (level 0) from hour 0 to 6, then at 0.5 until hour 10, then fully back. How many hours out?

Answer
6 × 1 + 4 × 0.5 = 8 hours out.

10. Travel between jobs
Drill
One crew sets out from the base, on a straight road. Facility B, on the road 8 km east, needs 1 h of work. Line L has two failure points, each 1 km off the road, at 12 km and 14 km east, each needing 6 h. Both jobs are in the same priority group. When does the crew finish line L?

Answer
B first, the shorter trip: done at hour 3.2. To L's first point: 4 km by road and 1 km on foot, 0.6 h, so there at 3.8; prepared at 5.8; done at 11.8. To the second point: 2

(see the other side)